

Jessen Narainsamy MOONEYAN

Senior Design Manager

PROFILE

Civil / Geotechnical Engineer with 19 years of experience in design and construction management and particular expertise in ground engineering within the highways and railway industries. Has worked mostly in the design consultancy sector managing geotechnical risk on ground-related projects: desk studies, scoping and specifying GI works, technical inspection and coordination of GI and ground works, scheduling soil tests, and producing interpretative and design reports, earthworks specifications and drawings. Has delivered temporary and permanent geotechnical design solutions, including emergency remedial works, on major schemes such as HS2, Network Rail, Birmingham Airport runway extension, Longbridge and Bristol Emersons Green developments, and slope stabilisation for Telford & Wrekin Council and Highways England.

CORE SKILLS

- Geotechnical design – foundations, settlement, seepage, slope stability, retaining walls, soil nailing, piling platforms
- Ground investigation – scoping, specification, supervision, scheduling of testing, interpretative & design reports (GIR/GDR)
- Earthworks – MCHW Series 600 specifications / Earthworks inspection and testing / Landfill CQA
- DMRB Volume 4 / UK Specification for GI / BS 1377 / NR/L3/CIV/071 / NR/L2/CIV/005 / BS EN 1997-1 & 1997-2
- Slope/W / LimitState GEO / WALLAP (sheet piling, king post, contiguous and secant walls) / HoleBASE SI / AutoCAD
- Temporary works – geotechnical design and coordination / Emergency remedial works / Track drainage design
- Work order and package management / Fee proposals / Client, contractor and stakeholder liaison

PROFESSIONAL EXPERIENCE

❖ TIFAND LIMITED — (March 2024 – Present)

Senior Design Manager

- Deliver geotechnical design and advice for highways, rail and civil infrastructure schemes, from desk study through to detailed design and construction support.
- Scope, specify and supervise ground investigation works, and schedule laboratory testing to BS 1377 and the UK Specification for Ground Investigation.

- Interpret ground investigation data to derive geotechnical design parameters and produce Ground Investigation and Geotechnical Design Reports.
- Design foundations, retaining walls, slope stabilisation, soil nailing, piling platforms and earthworks to Eurocode 7, DMRB and Network Rail standards.
- Undertake geotechnical analysis using Slope/W, LimitState GEO, WALLAP and HoleBASE SI.
- Provide geotechnical input to TIFAND temporary works designs and assess geotechnical risk on live sites.
- Prepare earthworks specifications (MCHW Series 600) and construction drawings.
- Carry out site inspections, monitoring and reporting, liaising with clients, contractors and Temporary Works Coordinators.
- Manage geotechnical work packages to programme and budget, including fee proposals and resource coordination.

❖ COFFEY GEOTECHNICS LTD (A TETRA TECH COMPANY) — *(February 2023 – March 2024)*

Lead Geotechnical Design Representative – Client: EKFB

- Lead design representative providing site-based consulting services to the EKFB temporary works coordination team on behalf of Coffey Geotechnics.
- Geotechnical inspections, risk assessments, review and reporting of a 30 km linear asset of HS2 from Banbury to Coventry, including embankments, cuttings, tunnels, culverts, overbridges, viaducts and earthworks.

❖ LEEKE ASSOCIATES LTD — *(December 2020 – February 2023)*

Senior Design Engineer – Client: Network Rail

- Cowarden Drainage (2022): designed track drainage flood alleviation along a 1 km length of cess, connecting to existing drains and outfalling to a culvert, including catchment assessment and pipe capacity.
- Arley Cutting, Nuneaton (2021): designed a soil nailing solution with mesh and mat, construction sequencing and a temporary mid-slope catch fence to protect live rail traffic.
- Brent Cross, London (2021): designed a temporary piling platform to BRE 470, evaluating capping performance with and without geogrid reinforcement.
- Faintree Pools, Bridgnorth (2021): assessed a flood defence/dam for slope stability, permeability, piping and uplift failure, and appraised a proposed additional dam.
- Ockley Embankment, Surrey (2020): emergency remedial works – designed a toe retaining wall using sheet pile and king post wall solutions with anchored waler beams on shallow piles.

❖ WSP — *(June 2017 – November 2020)*

Geotechnical Engineer – Client: Northamptonshire County Council

Work Order Manager – St James Mill Link Road, Northampton (2017)

120 m proposed link road underlain by landfill, crossing a redundant ditch and a disused railway.

- Assessed earthworks for removal of landfill waste and replacement with compacted engineered fill, settlement of underlying cohesive deposits, and uplift of culvert boxes with foundation options.

- Produced detailed design calculations, construction drawings, Series 600 earthworks specifications and the GDR.
- Prepared the fee proposal under framework rates, allocating resources and coordinating input from other disciplines.
- Appointed Work Order Manager on fee acceptance; coordinated resources, progress and budget, delivering the extended work package on time and within budget.

❖ CH2M — *(March 2013 – May 2017)*

Geotechnical Engineer – Clients: Highways Agency & St Modwen

M1 Soil Nailing, Northampton (2015)

- Inspected and monitored the installation of soil nails to stabilise an M1 embankment slope, resolving constraints including test nail grout set, tree stumps, service clashes, an underlying culvert and mesh installation.
- Created a live spreadsheet of all activities and issues to resolve problems efficiently, and produced the GFR for the Highways Agency.

Earthworks, Longbridge Development (2013)

- Carried out twice-weekly inspections of phased regeneration earthworks on contaminated sites to ensure compliance with specification.
- Collaborated with the main contractor to resolve site issues and reviewed the earthworks testing regime against specified requirements.

❖ JACOBS — *(July 2007 – February 2013)*

Graduate to Assistant Geotechnical Engineer – Client: Argyll & Bute Council

Assistant Project Manager – Gartbreck Landfill, Isle of Islay (2012)

- CQA Engineer on site, ensuring quality assurance of manufacture, testing, transport, handling and installation of earthworks (including lime mix), BES, liner materials (GCL, HDPE, GDL, filter geotextile) and leachate drainage.
- Monitored and recorded contractor activities, reviewed test results and recorded deviations against Series 600 and liner system specifications.
- Created spreadsheets and graphs for stepwise monitoring, helping the contractor remediate shortfalls, and prepared the validation report.

EDUCATION & TRAINING

- MSc (Hons) Geotechnical Engineering, Merit – University of Birmingham (2014)
- MEng (Hons) Civil Engineering, 2:1 – University of Birmingham (2007)
- ICE course in Civil Engineering Law and Contract Management (2015)
- Temporary Works Coordinator Training Course
- CSCS / Safe Supervision of Geotechnical Sites / Working on Highways / Airport General Security

LANGUAGES

- Fluent English
- Fluent French